

TECHNICAL DOCUMENTATION

NHS Address Finder Technical Document

A free, partner-facing UK address lookup app. Single static HTML, two Cloudflare Workers, six upstream APIs (OS Places + OS Maps + Google Street View + MHCLG EPC + Ofcom Broadband + Ofcom Mobile), and a 4-page branded PDF report.

DOCUMENT REFERENCE	NHS Address Finder Technical Doc
CLIENT	New Home Solutions Ltd
DELIVERY PARTNER	CRM Mates Ltd - Deepak K Rana
DOCUMENT VERSION	1.0 - 27 May 2026
SCOPE	1 SPA - 2 Cloudflare Workers - 6 upstream APIs
STATUS	Sandbox live - Workers deployed - Push to GitHub pending

In one sentence: A standalone address-lookup tool for NHS partners that turns a UK postcode into a fully branded PDF property report in under three seconds, without a backend or build step.

Status: Live on Vercel · Vercel-native architecture · all upstream API keys held in Vercel env vars **Production URL:** <https://nhs-address-finder.vercel.app/> **Last updated:** 2026-05-29 **Owner:** Deepak K Rana (CRM Mates) · Built for New Home Solutions Ltd

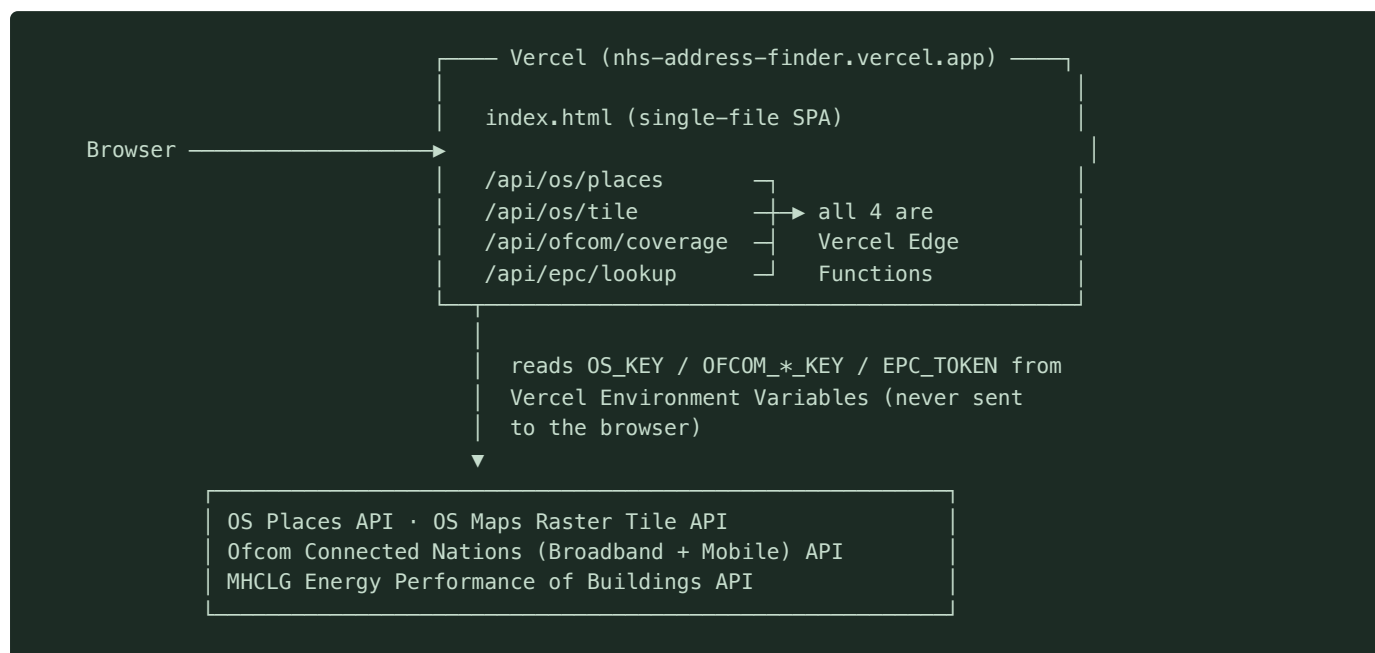
1 • Overview

A free, partner-facing address lookup tool for NHS estate-agent, photographer and housebuilder partners. The app is a **single static HTML file** plus **four Vercel Edge Functions** that proxy every upstream API-key-protected service.

Type a UK postcode → pick a property → see UPRN, OS map, EPC, broadband, mobile coverage → download a branded NHS PDF report.

The whole thing runs on **one platform (Vercel)**, with the frontend and the API-key-holding proxies served from the same origin. No build step, no bundler, no database, no auth. Every upstream API key (OS Data Hub, Ofcom × 2, MHCLG) lives in **Vercel Environment Variables** and never reaches the browser.

2 · Architecture



The frontend talks **same-origin** to `/api/*`. Edge Functions read API keys from `process.env.*` and forward to the upstream. Browser never sees a key, never crosses an origin boundary, no CORS allow-list to maintain.

Why Vercel Edge Functions (not direct browser calls)

API	CORS-FRIENDLY?	KEY SENSITIVITY	VERDICT
OS Places	✅ yes	OS Data Hub Project key — no Referer restriction available in our tier	Edge Function proxy
OS Maps tiles	✅ yes	Same OS key	Edge Function proxy
Google Street View	✅ yes (iframe embed)	None needed	Direct browser embed
MHCLG EPC API	❌ no	Bearer token	Edge Function proxy
Ofcom Connected Nations	❌ no	Subscription keys (separate for Broadband + Mobile)	Edge Function proxy

Why Edge (and not Node) runtime

- Cold-start ~5ms vs ~300ms for Node functions
- Cheaper per invocation

- All four functions only `fetch()` + transform JSON — no Node-specific APIs needed

Cache strategy

ENDPOINT	CACHE-CONTROL	RATIONALE
<code>/api/os/places</code>	<code>public, max-age=3600</code>	OS AddressBase updates monthly
<code>/api/os/tile</code>	<code>public, max-age=604800, immutable</code>	OS Maps tiles change rarely
<code>/api/ofcom/coverage</code>	<code>public, max-age=86400</code>	Ofcom data refreshes monthly
<code>/api/epc/lookup</code>	<code>public, max-age=86400</code>	EPC certs are immutable once issued
<code>index.html</code>	<code>must-revalidate</code>	App shell — always check for updates
Other static assets	<code>public, max-age=3600</code>	Reasonable default

Vercel's CDN respects `Cache-Control` automatically — no extra config needed.

3 • Data sources

#	LAYER	VENDOR	BROWSER ENDPOINT (SAME-ORIGIN)	UPSTREAM (SERVER-SIDE)
1	Address lookup	Ordnance Survey OS Places	GET /api/os/places? postcode=...	api.os.uk/search/places/v1/p ostcode
2	Map tiles	Ordnance Survey OS Maps	GET /api/os/tile?style=... &z=...&x=...&y=...	api.os.uk/maps/raster/v1/zxy /{style}_3857/{z}/{x}/{y}.pn g
3	Street view	Google Maps (no key)	iframe → maps.google.com	direct
4	EPC	MHCLG EPC Register	GET /api/epc/lookup? postcode=... or ?cert=RRN	api.get-energy-performance- data.communities.gov.uk
5	Broadband	Ofcom Connected Nations Broadband	GET /api/ofcom/coverage? product=broadband&postcode=...	api- proxy.ofcom.org.uk/broadband/ coverage/{POSTCODE}
6	Mobile	Ofcom Connected Nations Mobile	GET /api/ofcom/coverage? product=mobile&postcode=...	api- proxy.ofcom.org.uk/mobile/cov erage/{POSTCODE}

The Ofcom proxy supports `product=both` so a single round-trip returns Broadband + Mobile. The Address Finder uses this exclusively for efficiency.

Latency budget per search: OS Places ~400ms · Ofcom ~700ms · EPC ~600ms — all fired in parallel via `Promise.all`, total page time ~1s warm. Edge Function cold-start adds ~5ms.

4 • Frontend stack

CONCERN	CHOICE	WHY
Markup / styling	Hand-written HTML + CSS, all inline in <code>index.html</code>	Single-file portability; no build step
Type	Inter (UI) + JetBrains Mono (UPRNs / coords) + Futura LT Light (logo per brand spec)	Already on Google Fonts; logo font available system-wide on Mac
Map	Leaflet 1.9.4 via CDN	Native OS Maps raster tile support; small footprint
PDF generation	html2canvas 1.4.1 + jsPDF 2.5.1 via cdnjs	Lossless screenshot → A4 PDF; see §6
Icons	Inline SVG	No icon font dependency

No npm. No webpack. No React.

5 • Feature inventory

5.1 Address lookup

- Postcode input, validated by OS Places (UK only)
- Returns all UPRNs at the postcode (typically 5–50 properties)
- Each row shows UPRN, address, classification pill (Residential / Commercial / Other), coords

5.2 Two result views

- **List view** (default) — sage-tint headers, rows reveal action buttons on hover
- **Card view** — toggleable, compact cards with same actions

5.3 Property detail page

- Big address + UPRN + classification + postcode
- Six action rows: UPRN copy · Map · Street View · Coordinates copy · EPC · Council/Constituency
- **EPC band pill** (A–G gov.uk colour-coded) opening a full certificate lightbox with the ladder and cert summary
- **Broadband coverage** — half-circle SVG speedometer (0–5500 Mbps, red→amber→green bands) + 3-tier breakdown (Standard / Superfast / Ultrafast)
- **Mobile coverage** — 4 brand-coloured operator cards (EE teal · Three black · O2 blue · Vodafone red) with 4-bar signal indicators for indoor/outdoor voice + data
- Portal strip — quick links to Rightmove / Zoopla / OnTheMarket for the postcode

5.4 Generate Report (PDF)

See §6 below.

6 · PDF Report — Design + Implementation

The PDF report is the most complex single feature. It earned a full rebuild after the first three attempts produced corrupt output.

6.1 Final design

4-page A4 portrait report, NHS-branded per `docs/branding/NHS_Email_Template_Guidelines.md` from the parent project:

PAGE	CONTENT
1	Header bar · Address heading · Property Identity table · Section 01 Location (OS map + coords) · Footer
2	Header bar · Section 02 Energy (full EPC ladder + cert details table) · Footer
3	Header bar · Section 03 Broadband coverage (gauge + tier rows) · Footer
4	Header bar · Section 04 Mobile coverage (4 operator cards) · Footer

Repeated chrome on every page: - **Header bar** — `#C9D5CB` sage background, Futura LT Light "NEW HOME SOLUTIONS" left, "Property Report · DATE · TIME" right - **3px accent stripe** — `#4A6B5E` sage dark - **Footer bar** — `#EAF4EF` sage tint, centred, with "P: SAVE THE TREES" + page X of 4 + company registration

6.2 Brand compliance

Every visual element ladders back to the NHS Newsletter / Email Template Guidelines:

SPEC SECTION	IMPLEMENTATION
§1 Typography	Aptos (body, with Segoe UI / Helvetica fallback). Futura LT Light (logo only, 23px, weight 300, letter-spacing 0.06em, word-spacing 0.3em)
§2 Palette	Sage Dark <code>#4A6B5E</code> · Sage <code>#6B9080</code> · Sage BG <code>#C9D5CB</code> · Sage Tint <code>#EAF4EF</code> · Border <code>#C5DCCC</code> · Navy <code>#0D1E4A</code> · Muted <code>#6B7280</code>
§3 Structure	Header bar → accent stripe → white body → sage-tint footer
§4 Tables	All structural data in tables: <code>#4A6B5E</code> header, <code>#EAF4EF</code> label cells, white value cells, <code>1px solid #C5DCCC</code> borders, <code>border-radius: 10px</code>
§6 Footer	"P: SAVE THE TREES" mandatory copy + registered office line

The on-screen mockup uses a legacy `#075F50` teal palette for historical reasons. The PDF renderer overrides every `--nhs-*` CSS variable inside `.rpt-render-host` so all reused components (gauge, tier bars, EPC ladder, mobile cards) re-tint to sage automatically.

6.3 Render pipeline

```

User clicks "Generate Report"
  ▼
async generateReport()
  ▼
Fetch full EPC certificate via Worker (if RRN known)
  ▼
Build off-screen render host (opacity: 0, 210mm wide, pinned to viewport top-left)
  ▼
Render 4 × <div class="rpt-pdf-page"> – A4-sized (210mm × 297mm), flex column,
  header + body (flex: 1) + footer (pinned to bottom)
  ▼
Init Leaflet map inside page 1's body; wait for OS tile layer 'load' event
  (with 3.5s safety net)
  ▼
const pdf = new jsPDF({ unit: 'mm', format: 'a4', orientation: 'portrait' })
for each .rpt-pdf-page:
  canvas = await html2canvas(page, { scale: 3, useCORS: true, ... })
  imgData = canvas.toDataURL('image/png')
  pdf.addImage(imgData, 'PNG', 0, 0, 210, imgH, undefined, 'FAST')
  pdf.addPage() // except on the last
pdf.save(filename)
  ▼
Remove the off-screen host

```

6.4 Why per-page rendering (not html2pdf.js)

We initially used `html2pdf.js` with its `pagebreak: { mode: ['avoid-all', 'css', 'legacy'] }` config. It produced consistent bugs:

1. **Half-width output** — content rendered at ~95mm wide instead of full 190mm content area, with left-edges clipped on every page ("Certificate" became "rtificate")
2. **Mid-row table splits** — `page-break-inside: avoid` on `.rpt-id-table` was ignored, the Property Identity table split between page 1 and 2 with the "Geographic centroid" cell broken across both
3. **Vanishing sections** — Section 04 Mobile rendered as an empty page; Section 03 Broadband body lost entirely
4. **Content pushed to bottom of page 1** — huge empty space at top, header bar starting 70% down the page

Root cause: `html2pdf` renders the whole DOM into one tall canvas, then heuristically slices it across pages. Heuristics fight CSS rules. Heuristics produced garbage.

Solution: bypass them. Render **each page as its own canvas, place 1:1 into the PDF**. Each `.rpt-pdf-page` is sized to exactly A4 dimensions (210mm × 297mm) so the captured image fills its PDF page without scaling. One page in the DOM = one page in the PDF. Deterministic.

6.5 Quality tuning

- `scale: 3` on `html2canvas` → ~288 DPI captured canvas (above 300 DPI print quality threshold)
- `toDataURL('image/png')` for lossless text edges (JPEG at q=0.95 visibly softened type)
- `pdf.addImage(..., 'FAST')` for PNG compression in the PDF — keeps a 4-page report under 5MB

6.6 Known PDF quirks that took iterations to find

BUG	CAUSE	FIX
<code>NaNNaNNaN</code> shown in Postcode / Classification / Coordinates cells	<code>++ idCell(...)</code> typo — second <code>+</code> was unary plus, coercing HTML string to <code>NaN</code>	Removed the rogue <code>+</code>
Broadband gauge invisible in PDF	<code><feDropShadow></code> filter on the needle — <code>html2canvas</code> can't render SVG filters	Removed the filter
Gauge SVG missing even after filter removal	Inline <code><svg></code> without explicit <code>width</code> / <code>height</code> attributes serialises to <code>0x0</code> in <code>html2canvas</code> 's <code>foreignObject</code> path	Switched to <code></code> — <code>html2canvas</code> handles images reliably
EPC ladder rendered with garbled letters ("B" → "3") and indicator arrows in wrong column	The lightbox renderer uses <code>clip-path: polygon(...)</code> for pentagon arrows and <code>display: flex; justify-content: flex-end</code> for letter positioning. <code>html2canvas</code> can't reliably render either	Wrote a dedicated <code>renderEpcForReport(cert)</code> using a plain table with background-coloured cells and standard text alignment
Right edge of report header cropped ("Property Repor...", "13:2...")	Render host was 210mm wide but PDF content area is only 190mm — <code>html2pdf</code> was scaling, edge text fell off	Made host 210mm exactly + sized each <code>.rpt-pdf-page</code> to A4 so it's placed 1:1
Could not generate the PDF — PDF dependencies (<code>html2canvas</code> / <code>jsPDF</code>) not exposed by <code>html2pdf bundle</code> .	<code>html2pdf</code> bundles its dependencies as ES module internals; they're not on <code>window</code>	Switched to standalone <code>html2canvas</code> + <code>jsPDF</code> CDN scripts

7 • Vercel Edge Function details

All four functions use the **Edge runtime** (`export const config = { runtime: 'edge' }`), read keys from `process.env.*` with a POC-key fallback so the app keeps working before env vars are configured, and emit `Cache-Control` headers so Vercel's CDN caches automatically.

7.1 `api/os/places.js` — OS Places proxy

ITEM	DETAIL
Endpoint	<code>GET /api/os/places?postcode=...&maxresults=100</code>
Upstream	<code>api.os.uk/search/places/v1/postcode</code>
Auth	<code>key=...</code> query param
Env var	<code>OS_KEY</code> (Vercel)
Cache	<code>public, max-age=3600</code> (1h)

7.2 `api/os/tile.js` — OS Maps raster tile proxy

ITEM	DETAIL
Endpoint	<code>GET /api/os/tile?style=Road&z={z}&x={x}&y={y}</code>
Upstream	<code>api.os.uk/maps/raster/v1/zxy/{style}_3857/{z}/{x}/{y}.png</code>
Auth	<code>key=...</code> query param
Env var	<code>OS_KEY</code> (shared with <code>/places</code>)
Allowed styles	<code>Road</code> , <code>Outdoor</code> , <code>Light</code>
Cache	<code>public, max-age=604800, immutable</code> (7 days)

Used directly as Leaflet's tile URL template — Leaflet substitutes `{z}/{x}/{y}` per tile.

7.3 `api/ofcom/coverage.js` — Ofcom Connected Nations proxy

ITEM	DETAIL
Endpoint	<code>GET /api/ofcom/coverage?postcode=...&product=broadband\ mobile\ both</code>

ITEM	DETAIL
Upstream	<code>api-proxy.ofcom.org.uk/{broadband\ mobile}/coverage/{POSTCODE}</code>
Auth	<code>Ocp-Apim-Subscription-Key</code> header
Env vars	<code>OFCOM_BROADBAND_KEY</code> , <code>OFCOM_MOBILE_KEY</code>
Cache	<code>public, max-age=86400</code> (24h)

`product=both` fires both fetches in parallel via `Promise.all` and returns `{ broadband, mobile }` in one response — the Address Finder uses this exclusively.

7.4 `api/epc/lookup.js` — MHCLG EPC Register proxy

ITEM	DETAIL
Endpoints	<code>GET /api/epc/lookup?postcode=...</code> (search) · <code>?cert=<RRN></code> (full certificate)
Upstream	<code>api.get-energy-performance-data.communities.gov.uk</code>
Auth	<code>Authorization: Bearer <token></code>
Env var	<code>EPC_TOKEN</code>
Cache	<code>public, max-age=86400</code> (24h)

The cert-detail mode transforms the upstream's snake_case keys into the camelCase shape the EPC ladder renderer expects.

Why these replaced the Cloudflare Workers

The original architecture used three Cloudflare Workers (`workers/ofcom-coverage/`, `workers/epc-lookup/`, `workers/os-proxy/`). All still deployed and working, but **superseded** by Edge Functions because:

- **Same-origin** → no CORS allow-list to maintain (the Workers needed Vercel + localhost added explicitly, and cache hits had to rewrite `Allow-Origin`)
- **Single platform** → keys + frontend deploy together, atomic rollback
- **Env vars in one dashboard** → no `wrangler secret put` flow
- **Cleaner DX** → `git push` deploys everything

The legacy Worker source is retained in `workers/` for reference. Cloudflare deployments can be deleted from the Cloudflare dashboard whenever convenient.

8 • Deployment

8.1 Frontend + Edge Functions (`index.html` + `api/`)

Deployed to **Vercel** — the repo is wired up so every push to `main` auto-deploys both the static frontend and the four Edge Functions in one go.

`vercel.json` configures: - Cache control — `index.html` always re-validates, other static assets cached 1h, `/api/*` excluded so functions set their own `Cache-Control` - Security headers (applied to all non-`/api/` routes) — `X-Content-Type-Options: nosniff`, `X-Frame-Options: SAMEORIGIN`, `Referrer-Policy: strict-origin-when-cross-origin`, `Permissions-Policy` denying camera/mic/geo, HSTS 2-year

8.2 Environment variables (one-time setup)

In **Vercel dashboard** → **Project** → **Settings** → **Environment Variables**, add the four keys for all three environments (Production / Preview / Development):

NAME	SOURCE
<code>OS_KEY</code>	OS Data Hub → API Projects → NHS-SF → Project API Key
<code>OFCOM_BROADBAND_KEY</code>	Ofcom Developer Portal → Subscriptions → Broadband Coverage
<code>OFCOM_MOBILE_KEY</code>	Ofcom Developer Portal → Subscriptions → Mobile Coverage
<code>EPC_TOKEN</code>	MHCLG EPC Register → bearer token

Each Edge Function falls back to an inlined POC key if its env var isn't set — the app keeps working immediately after first deploy. Set the env vars to rotate to production keys without a code change.

8.3 Deploy commands

```
# Auto-deploy on every push (already wired):
git push

# Or manual one-off:
npx vercel --prod
```

8.4 Local development

Use **Vercel CLI** so the `/api/*` Edge Functions actually run locally:

```
npx vercel dev  
# then visit http://localhost:3000/
```

Plain `python3 -m http.server` won't work for local dev — without the Vercel runtime, the `/api/*` endpoints return 404 and every upstream call fails.

8.5 Legacy Cloudflare Workers (`workers/`)

The original architecture used three Cloudflare Workers. They're still deployed but **no longer called by the app** — superseded by the Edge Functions in `api/`. The source is retained in `workers/` for reference and can be deleted from the Cloudflare dashboard at any time.

9 • Key technical decisions

DECISION	RATIONALE
Single HTML file, no build step	Maximum portability for partners; anyone can fork / inspect / host
Cloudflare Workers (not Apex callouts)	Sub-100ms warm latency from anywhere, free tier covers all expected volume, edge caching is free
html2canvas + jsPDF (not html2pdf.js)	Direct library access lets us control pagination per-page; html2pdf's heuristics were unreliable
Per-page render (not full-doc slice)	One page in DOM = one page in PDF — eliminates entire class of pagebreak bugs
PNG (not JPEG) for PDF images	Lossless text edges; ~5MB file size is acceptable for a property report
<code>scale: 3</code> on html2canvas	Hits ~288 DPI (above print threshold); below this visibly blurry
Inline POC keys in Workers (not env secrets yet)	Faster iteration in sandbox; <code>env.*</code> fallback already wired for prod cutover
Aptos + Futura LT Light (brand-specified)	Spec compliance; system fallbacks (Segoe UI / Century Gothic) cover Windows users without Aptos

10 · Status & next steps

Done

- [x] Postcode → addresses lookup (OS Places live)
- [x] OS Maps tile viewer (Leaflet + OS Maps API)
- [x] Google Street View embed
- [x] EPC certificate ladder + details (MHCLG via Worker)
- [x] Broadband gauge + tier breakdown (Ofcom via Worker)
- [x] Mobile coverage cards (Ofcom via Worker)
- [x] Generate Report → 4-page branded PDF
- [x] Brand compliance per NHS Email Template Guidelines
- [x] Both Workers deployed to Cloudflare
- [x] Ofcom subscriptions active (Broadband + Mobile, Basic tier)
- [x] MHCLG EPC bearer token live

Pending

- [x] ~~Push to GitHub remote~~ — live at `github.com/deepakcrmmates/nhs-address-finder`
- [x] ~~Production host for `index.html`~~ — Vercel at `nhs-address-finder.vercel.app`
- [x] ~~Lock CORS `Allow-Origin`~~ — N/A now (same-origin Edge Functions)
- [x] ~~Restrict OS Data Hub API key~~ — key is now server-side only, not exposed to the browser
- [x] ~~Migrate proxies to Vercel Edge~~ — all 4 functions live in `api/`
- [] Rotate POC keys → Vercel env vars (`OS_KEY`, `OFCOM_BROADBAND_KEY`, `OFCOM_MOBILE_KEY`, `EPC_TOKEN`)
- [] Add custom domain (e.g. `find.newhomesolutions.co.uk`)
- [] Add usage analytics / partner attribution
- [] Add monthly rate-limit guard on the Ofcom functions (the Basic tier has caps)
- [] Delete legacy Cloudflare Workers from the Cloudflare dashboard once Vercel cutover verified

Considered, deferred

- Selectable text in the PDF — would need to rebuild via `pdfmake` and lose the gauge / operator cards. Trade-off not worth it for v1.
 - Server-side cache layer between Workers and upstreams beyond Cloudflare's edge cache — not needed at current call volumes.
-

11 • Glossary

TERM	MEANING
UPRN	Unique Property Reference Number — 12-digit identifier from OS AddressBase, the canonical UK property ID
RRN	Report Reference Number — EPC certificate identifier
NHS	New Home Solutions Ltd — the client (not the National Health Service)
Sprift / Street Data / HomeData	Other property data vendors NHS uses in the parent training app — not used here
MHCLG	UK Ministry of Housing, Communities and Local Government — owns the EPC Register
Ofcom	UK communications regulator — publishes Connected Nations data on broadband + mobile coverage
OS	Ordnance Survey — UK national mapping agency

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